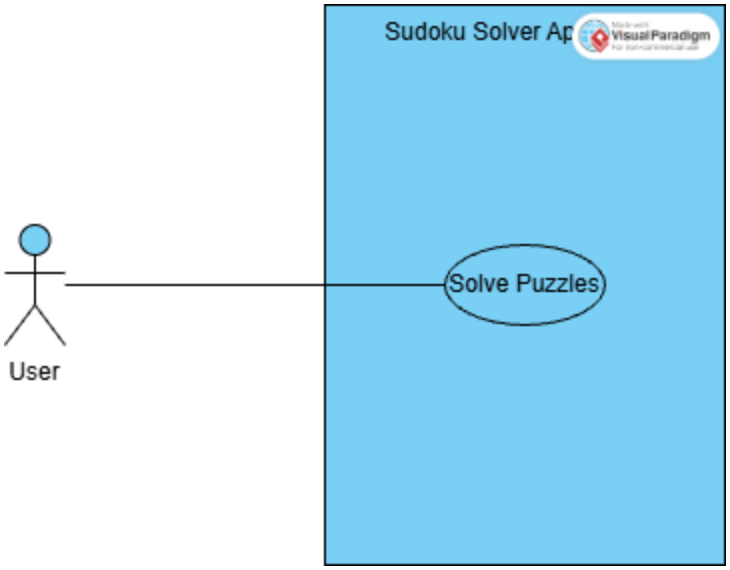


1. Identify/Discover a Use Case: Solve Sudoku Puzzles.
2. Brief Description: This use case allows a user to provide text files containing incomplete Sudoku grids to the system to calculate and display all possible valid mathematical solutions.
3. Step-by-Step Flow (Outline):

1. The User executes the program.
2. The System attempts to read the target puzzle text file.
3. The System breaks down the numbers and underscores into a mathematical grid.
4. The System displays the initial unsolved puzzle.
5. The System continuously searches for solutions using a backtracking algorithm.
6. The System prints all discovered solutions, or an error message if none exist.

4. Detailed Description: The use case starts when the User launches the executable. The System automatically selects a predefined list of text files (puzzle1.txt through puzzle5.txt). For each file, it opens a file stream, validates the contents, and maps the characters into a 9x9 internal integer matrix, converting underscores into zeros. Then, the system displays the original loaded board to the User. A recursive depth-first search algorithm tests numbers 1 through 9 in every empty cell, verifying row, column, and 3x3 block constraints. If a valid layout is found, the System prints the completed board and increases a solution counter. Once all possibilities are exhausted, the System either reports the total number of solutions found or outputs "No solution found" if the grid was invalid. The use case concludes when all files have been processed.